Empowering K-12 Students with Computational Creativity

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Abstract

Recent developments in Artificial Intelligence (AI) are not only transforming society but are also increasingly shaping educational contexts. While AI technologies offer new possibilities for rich learning experiences, there is growing consensus that students should not only use AI systems but also be able to understand and design them. Constructionist learning environments provide a promising foundation for this shift, enabling learners to engage hands-on with AI by constructing meaningful artifacts. One particularly suitable domain for this is Computational Creativity (CC), which focuses on systems that generate novel outputs using AI techniques. In this paper, we introduce the Constructionist Computational Creativity (CCC) model, which aims to integrate CC into K–12 education in a way that fosters both creative expression and AI competencies. The model was developed through a synthesis of CC theory and constructionist pedagogy and was refined through an exploratory study with pre-service Computer Science teachers. Findings from this study show that engaging learners in the development of creative AI systems supports a deeper understanding of AI concepts, enhances computational thinking, and promotes reflection on creativity across domains. The CCC model thus offers a structured approach to integrating AI education into creative learning processes.